

## CS5787: Exercises 2

Python Notebook and README practical section here

Avinash Nair  
an586

Jasper Levy  
jl4537

### 1 Theory: Question 1 [10 pts]

- a. One way to deal with the issue of variable-length input sequences is to pad all the sequences to the length of the longest sequences in a given minibatch. This is done by adding zeros to the end of shorter sequences. A masking mechanism can then be used to ignore the zeros during processing.
- b. The same approach for input sequences can be applied for output sequences. The sequences can be padded to match the longest length sequence. Loss computation would change, as a mask would need to be applied to make sure that pad tokens aren't included in loss calculations.

### 2 Theory: Question 2 [10 pts]

GRU's have a simpler structure than LSTM's. LSTM's have 3 gates: input, output, and forget. GRU's only have 2 gates: update and reset. This simpler architecture can lead to better performance on smaller datasets.

As a result of its simpler architecture, GRU's have fewer parameters than LSTM's. This can lead to faster training times for GRU's.

### 3 Theory: Question 3 [10 pts]

There are 4 relevant LSTM cell equations: Input gate, Output gate, Forget gate, and Input Node. All of these equations include weights for the input and hidden state. As such, we can calculate the total number of parameters as  $4 \times ((\text{input size} \times \text{hidden size}) + (\text{hidden size} \times \text{hidden size}) + \text{hidden size})$ . Plugging in 200 for input size and hidden size, we get  $4 \times ((200 \times 200) + (200 \times 200) + 200) = 320,800$ .

Thus, this LSTM cell has 320,800 parameters.

### 4 Theory: Question 4 [20 pts]

(a)  $\frac{\partial \epsilon^{(2)}}{\partial W_{xz}}$

Using update gate:

$$z^{(2)} = \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})$$

Gradient:

$$\frac{\partial z^{(2)}}{\partial W_{xz}} = \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot \left(1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})\right) \cdot x^{(2)}$$

Thus, the gradient of the loss with respect to  $W_{xz}$  is:

$$\frac{\partial \epsilon^{(2)}}{\partial W_{xz}} = \frac{\partial \epsilon^{(2)}}{\partial z^{(2)}} \cdot \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot \left(1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})\right) \cdot x^{(2)}$$

Need to find:

$$\frac{\partial \epsilon^{(2)}}{\partial z^{(2)}}$$

Using equation for:

$$h_{(t)}$$

$$h^{(2)} = (1 - z^{(2)}) \odot h^{(1)} + z^{(2)} \odot g^{(2)}$$

Differentiate  $h^{(2)}$  with respect to  $z^{(2)}$ :

$$\frac{\partial h^{(2)}}{\partial z^{(2)}} = g^{(2)} - h^{(1)}$$

Applying chain rule:

$$\frac{\partial \epsilon^{(2)}}{\partial z^{(2)}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot (g^{(2)} - h^{(1)})$$

Substitute the expression for  $\frac{\partial h^{(2)}}{\partial z^{(2)}}$ :

$$\frac{\partial \epsilon^{(2)}}{\partial z^{(2)}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot (g^{(2)} - h^{(1)})$$

Now that we have  $\frac{\partial \epsilon^{(2)}}{\partial z^{(2)}}$ , we can plug it into the expression for  $\frac{\partial \epsilon^{(2)}}{\partial W_{xz}}$ :

$$\frac{\partial \epsilon^{(2)}}{\partial W_{xz}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot (g^{(2)} - h^{(1)}) \cdot \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot (1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})) \cdot x^{(2)}$$

(b)  $\frac{\partial \epsilon^{(2)}}{\partial W_{hz}}$

Similarly, the gradient of  $z^{(2)}$  with respect to  $W_{hz}$  is:

$$\frac{\partial z^{(2)}}{\partial W_{hz}} = \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot (1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})) \cdot h^{(1)}$$

Thus, the gradient of the loss with respect to  $W_{hz}$  is:

$$\begin{aligned} \frac{\partial \epsilon^{(2)}}{\partial W_{hz}} &= \frac{\partial \epsilon^{(2)}}{\partial z^{(2)}} \cdot \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot (1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})) \cdot h^{(1)} \\ &= \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot (g^{(2)} - h^{(1)}) \cdot \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)}) \cdot (1 - \sigma(W_{xz}x^{(2)} + W_{hz}h^{(1)})) \cdot h^{(1)} \end{aligned}$$

(c)  $\frac{\partial \epsilon^{(2)}}{\partial W_{xg}}$

Candidate hidden state equation:

$$g^{(2)} = \tanh(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})$$

The gradient of  $g^{(2)}$  with respect to  $W_{xg}$  is:

$$\frac{\partial g^{(2)}}{\partial W_{xg}} = (1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})) \cdot x^{(2)}$$

Thus, the gradient of the loss with respect to  $W_{xg}$  is:

$$\frac{\partial \epsilon^{(2)}}{\partial W_{xg}} = \frac{\partial \epsilon^{(2)}}{\partial g^{(2)}} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \cdot x^{(2)}$$

Revisiting final hidden state:

$$h^{(2)} = (1 - z^{(2)}) \odot h^{(1)} + z^{(2)} \odot g^{(2)}$$

Differentiate  $h^{(2)}$  with respect to  $g^{(2)}$ :

$$\frac{\partial h^{(2)}}{\partial g^{(2)}} = z^{(2)}$$

Apply the chain rule:

$$\frac{\partial \epsilon^{(2)}}{\partial g^{(2)}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot \frac{\partial h^{(2)}}{\partial g^{(2)}}$$

Substitute  $\frac{\partial h^{(2)}}{\partial g^{(2)}} = z^{(2)}$ :

$$\frac{\partial \epsilon^{(2)}}{\partial g^{(2)}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)}$$

Plug in to get final gradient:

$$\frac{\partial \epsilon^{(2)}}{\partial W_{xg}} = \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \cdot x^{(2)}$$

(d)  $\frac{\partial \epsilon^{(2)}}{\partial W_{hg}}$

Similarly, for  $W_{hg}$ :

$$\frac{\partial g^{(2)}}{\partial W_{hg}} = \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \cdot r^{(2)} \cdot h^{(1)}$$

Gradient of the loss with respect to  $W_{hg}$  is:

$$\begin{aligned} \frac{\partial \epsilon^{(2)}}{\partial W_{hg}} &= \frac{\partial \epsilon^{(2)}}{\partial g^{(2)}} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \cdot r^{(2)} \cdot h^{(1)} \\ &= \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \cdot r^{(2)} \cdot h^{(1)} \end{aligned}$$

(e)  $\frac{\partial \epsilon^{(2)}}{\partial W_{xr}}$

From the reset gate equation:

$$r^{(2)} = \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})$$

The gradient of  $r^{(2)}$  with respect to  $W_{xr}$  is:

$$\frac{\partial r^{(2)}}{\partial W_{xr}} = \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot x^{(2)}$$

Thus, the gradient of the loss with respect to  $W_{xr}$  is:

$$\frac{\partial \epsilon^{(2)}}{\partial W_{xr}} = \frac{\partial \epsilon^{(2)}}{\partial r^{(2)}} \cdot \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot x^{(2)}$$

We need to find

$$\begin{aligned} & \frac{\partial \epsilon^{(2)}}{\partial r^{(2)}} \\ \frac{\partial \epsilon^{(2)}}{\partial r^{(2)}} &= \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)} \cdot (1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})) \odot W_{hg}h^{(1)} \end{aligned}$$

Finally:

$$\begin{aligned} \frac{\partial \epsilon^{(2)}}{\partial W_{xr}} &= \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \\ & \quad \odot W_{hg}h^{(1)} \cdot \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \\ & \quad \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot x^{(2)} \end{aligned}$$

(f)  $\frac{\partial \epsilon^{(2)}}{\partial W_{hr}}$

Similarly, the gradient of  $r^{(2)}$  with respect to  $W_{hr}$  is:

$$\frac{\partial r^{(2)}}{\partial W_{hr}} = \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot h^{(1)}$$

Thus, the gradient of the loss with respect to  $W_{hr}$  is:

$$\frac{\partial \epsilon^{(2)}}{\partial W_{hr}} = \frac{\partial \epsilon^{(2)}}{\partial r^{(2)}} \cdot \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot h^{(1)}$$

Plugging in for

$$\frac{\partial \epsilon^{(2)}}{\partial r^{(2)}}$$

We get:

$$\begin{aligned} \frac{\partial \epsilon^{(2)}}{\partial W_{hr}} &= \frac{\partial \epsilon^{(2)}}{\partial h^{(2)}} \cdot z^{(2)} \cdot \left(1 - \tanh^2(W_{xg}x^{(2)} + r^{(2)} \odot W_{hg}h^{(1)})\right) \\ & \quad \odot W_{hg}h^{(1)} \cdot \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)}) \\ & \quad \cdot \left(1 - \sigma(W_{xr}x^{(2)} + W_{hr}h^{(1)})\right) \cdot h^{(1)} \end{aligned}$$

## 5 Practical [50 pts]

### Link to folder

This includes the README with plots as well as the notebook with the code. You can access it here.

### Models

The models that we use in this practical are the LSTM and GRU RNNs. We used the "small" LSTM architecture as described in Zaremba et. al. (2015). The GRU model we used is very

similar to this LSTM model, and both model classes of PyTorch nn.module can be seen in the provided .ipynb notebook.

### **Conclusion and Discussion**

Out of all configurations of the models we tested, the GRU with dropout had the lowest training perplexity at 60.94. However, the lowest validation and test perplexity were both from the LSTM model with dropout, at 96.69 and 93.13, respectively.

For this LSTM model with dropout, the hyperparameters used were as follows:

1. **Dropout:** 0.33
2. **Learning Rate:** 6
3. **Learning Rate Decay Epoch:** 20
4. **Number of Epochs:** 30

The LSTM with dropout had higher train perplexity than its non-dropout LSTM counterpart, yet the model with dropout had lower perplexities for both validation and test sets. This makes sense generally, for the dropout provided regularization and led to a more generalized model as an end-result. The LSTM model had better performance (lower perplexities) on validation and test data. The GRU's worse performance on unseen test and validation data is possibly due to the simpler architecture of the GRU when compared to the LSTM, making it more sensitive to dropout and inhibiting learning to some extent.

Interestingly, the GRU model with dropout had lower train perplexity than its non-regularized GRU counterpart. Still, both GRU models had lower train perplexities than test and validation perplexities.

Notably, the GRU models required much lower starting learning rates for both the regularized and non-regularized models. Without these low learning rates, we initially saw extremely large perplexities in early epochs, suggesting that the gradients were more prone to explode. The LSTM model - particularly the one with dropout - was able to handle very high initial learning rates without gradient explosion.

Overall, while the GRU models were able to achieve low training perplexities with a smaller initial learning rate, the LSTM model with dropout generalized better on unseen test data.

Note: see README in drive folder for plots and perplexity table